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7.1 Introduction

The term 'Thermodynamics' means flow of heat and is derived from the Greek `Thermos' (heat) and `dynamics' (flow), In our daily life, we come across many useful reactions such as burning of fuel to produce heat energy, flow of electrons through circuit to produce electrical energy, metabolic reactions to produce the necessary energy for biological functions and so on. Thermodynamics, the study of the transformation of energy, explains all such processes quantitatively and allows us to make useful predictions.

In the 19th century, scientists tried to understand the underlying principles of steam engine which were already in operation, in order to improve their efficiency. The basic problem of the investigation was the transformation of heat into mechanical work. However, over time, the laws of thermodynamics were developed and helped to understand the process of steam engine. These laws have been used to deduce powerful mathematical relationships applicable to a broad range of processes.

Thermodynamics evaluates the macroscopic properties (heat, work) and their inter relationships. It deals with properties of systems in equilibrium and is independent of any theories or properties of the individual molecules which constitute the system.

The principles of thermodynamics are based on three laws of thermodynamics. The first two laws (First and second law) summarise the actual experience of inter conversion of different forms of energy. The third law deals with the calculation of entropy and the unattainability of absolute zero Kelvin. Thermodynamics carries high practical values but bears certain limitations. It is independent of atomic and molecular structure and reaction mechanism. The laws can be used to predict whether a particular reaction is feasible or not under a given set of conditions, but they cannot give the rate at which the reaction takes place. In other words, thermodynamics deals with equilibrium conditions quantitatively, but does not take into account the kinetic approach to the equilibrium state.

7.2 System and Surrounding

Before studying the laws of thermodynamics and their applications, it is important to understand the meaning of a few terms used frequently in thermodynamics.

System:

The universe is divided into two parts, the system and its surroundings. The system is the part of universe which is under thermodynamic consideration. It is separated from the rest of the universe by real or imaginary boundaries.

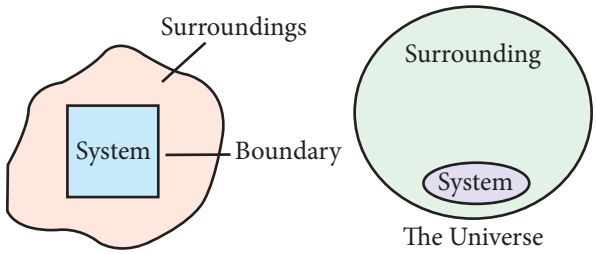

Figure 7.1 System, surrounding & boundary Homogeneous and heterogeneous systems